

BAHIA BLANCA, Argentina

WMO: 877500

Lat: 38.7147S

Lon: 62.1631W

Elev: 83

StdP: 100.33

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -2.2	(c) -0.8	(d) -7.3	(e) 2.0	(f) 6.7	(g) -5.8	(h) 2.3	(i) 6.2	(j) 19.2	(k) 13.2	(l) 16.5	(m) 11.7	(n) 3.3	(o) 320	(p) 0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 13.9	(c) 35.3	(d) 20.4	(e) 33.8	(f) 19.9	(g) 32.0	(h) 19.4	(i) 22.9	(j) 30.8	(k) 21.9	(l) 29.5	(m) 21.0	(n) 28.6	(o) 8.3	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.5	26.0	19.6	14.5	24.9	18.7	13.7	24.0	68.3	31.0	64.5	29.4	61.3	28.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.3	(b) 13.9	(c) 12.5	(e) -5.0	(f) 38.6	(g) 1.8	(h) 1.3	(i) -6.3	(j) 39.5	(k) -7.4	(l) 40.2	(m) -8.4	(n) 41.0	(o) -9.7	(p) 41.9		
			DB	WB												
					-5.6	25.0	1.8	1.0	-6.9	25.7	-8.0	26.3	-9.0	26.9	-10.3	27.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	23.5	22.2	19.7	15.5	12.0	8.9	8.4	10.0	12.2	15.2	18.8	21.9
	DBStd	6.37	3.50	3.82	3.85	3.69	3.61	3.22	3.50	3.76	3.83	3.66	3.76	3.81
	HDD10.0	221	0	0	0	2	18	57	75	46	19	3	0	0
	HDD18.3	1538	3	9	29	99	199	282	309	258	188	112	38	11
	CDD10.0	2285	419	341	300	168	81	25	24	47	84	163	264	368
	CDD18.3	561	163	117	72	15	3	0	0	1	3	13	53	121
	CDH23.3	6254	1877	1242	704	136	18	0	1	12	36	166	636	1426
	CDH26.7	2632	897	553	252	28	2	0	0	2	4	37	218	638
Wind	WSAvg	6.2	7.1	6.5	6.0	5.7	5.4	5.9	5.9	6.0	6.1	6.2	6.5	7.0
Precipitation	PrecAvg	624	62	68	79	58	40	25	29	30	47	71	64	64
	PrecMax	1093	169	249	261	211	112	101	90	138	122	227	241	211
	PrecMin	331	2	1	12	2	0	0	0	0	0	1	5	0
	PrecStd	141	40	54	52	48	31	24	25	29	32	48	39	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	36.7	34.9	29.5	26.0	20.0	21.1	24.6	26.8	30.1	34.1	37.0
		MCWB	20.9	21.3	21.1	18.6	17.8	13.4	13.6	15.3	15.7	18.2	18.9	21.0
	2%	DB	35.4	34.3	31.8	27.0	22.2	17.8	18.1	21.2	24.0	27.2	31.4	34.7
		MCWB	20.5	20.7	20.1	17.8	15.8	12.0	11.8	13.0	14.4	16.8	18.0	19.6
	5%	DB	33.6	32.2	29.5	24.5	19.9	16.1	16.2	19.0	21.9	25.0	29.1	32.5
		MCWB	20.1	20.0	19.3	16.7	14.5	10.9	10.7	11.9	13.2	15.8	17.3	18.9
	10%	DB	31.8	30.2	27.3	22.2	18.0	14.6	14.6	16.9	19.8	22.9	27.1	30.2
		MCWB	19.5	19.3	18.1	15.6	13.4	10.3	9.8	10.7	12.0	14.7	16.3	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.1	23.3	21.3	19.2	15.1	16.0	17.5	17.6	20.3	20.9	23.6
		MCDB	32.6	31.8	30.1	25.1	22.9	18.1	19.0	21.9	23.4	26.8	29.1	32.0
	2%	WB	22.7	22.7	21.7	19.2	17.3	13.5	13.4	14.2	15.8	18.4	19.6	21.8
		MCDB	31.0	30.2	28.7	24.2	20.7	16.0	16.1	19.0	21.8	24.5	27.7	30.5
	5%	WB	21.5	21.7	20.6	18.0	15.6	12.1	11.8	12.7	14.3	16.9	18.5	20.6
		MCDB	30.0	28.6	27.1	22.7	18.8	14.8	15.3	17.6	19.9	22.7	26.3	28.9
	10%	WB	20.6	20.8	19.5	16.7	14.1	10.8	10.3	11.3	13.0	15.7	17.5	19.4
		MCDB	28.7	27.4	25.3	21.0	17.0	13.9	13.6	15.8	18.6	21.4	24.8	27.4
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.1	12.5	11.7	10.4	10.3	10.6	11.3	12.3	12.4	13.6	14.1
		MCDBR	16.4	15.8	14.6	14.4	12.3	11.6	12.4	13.4	14.9	15.7	16.2	16.9
		MCWBR	6.4	6.2	6.3	7.2	7.0	7.1	7.5	7.5	7.8	7.8	6.9	6.7
	5% WB	MCDBR	14.4	13.0	12.7	11.8	10.4	9.7	10.6	11.9	13.1	13.5	14.0	14.5
		MCWBR	6.6	6.2	6.6	6.8	6.7	6.8	7.3	7.3	7.7	7.8	7.0	6.9
Clear-Sky Solar Irradiance	taub		0.394	0.381	0.366	0.360	0.336	0.321	0.323	0.366	0.399	0.380	0.380	0.394
	taud		2.347	2.388	2.435	2.404	2.442	2.484	2.465	2.314	2.223	2.326	2.349	2.322
	Ebn at Noon		937	924	896	831	789	774	797	808	844	915	945	942
	Edh at Noon		131	120	106	95	79	70	76	102	128	127	130	136
All-Sky Solar Radiation	RadAvg		7.83	6.69	5.26	3.65	2.37	1.98	2.14	2.95	4.23	5.54	7.14	8.00
	RadStd		0.30	0.27	0.23	0.34	0.19	0.19	0.18	0.33	0.26	0.40	0.38	0.43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	-72	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page